



PowerChain

DISASTER RELIEF | CRISIS CAPITAL NETWORK

Verified capital infrastructure for disaster response, recovery and measurable impact.

PWRC Utility Architecture · Visual Data Semantics · Solana Settlement · Evidence & Impact Fabric

Technical Whitepaper - Architecture & Data Edition
Version 1.0.0
28 August 2026

CORE PRINCIPLE

PWRC powers utility around the response. Stable assets fund the response. Identity and policy control authority. Cryptographic signers execute value movement. Evidence proves real-world outcomes.

Whitepaper contents

This edition aligns the whitepaper with the upgraded chart, connector, provenance and deployment-status system used across the PowerChain Crisis UI and architecture exports.

Section	Focus
01	Executive summary - capital to outcomes.
02	CCN architecture - bounded networks and truth layers.
03	Capital lifecycle - explicit financial states and reconciliation.
04	Fee model - contributor fees versus post-success commission.
05	PWRC utility - Power Units and authority separation.
06	Token factory - issuer profiles and mint-secret boundary.
07	Cross-chain - CCTP, CCIP and wPWRC/Sui fail-closed routes.
08	Security control plane - identity, policy, signer and reconciliation.
09	Community rewards - bounded evidence-based incentives.
10	Evidence & impact - proof beyond transaction confirmation.
11	UX/UI - public, operator, treasury and auditor experiences.
12	Nepal demo fixture - explicitly non-live product data.
13	Data/oracles - source provenance, freshness and provider state.
14	Visual data system - arrows, edge states, charts and accessibility.
15	Production readiness - mainnet release gates.
16	Strategic roadmap - activation sequence within v1.0.0.
17	Conclusion & references - governing invariants and sources.

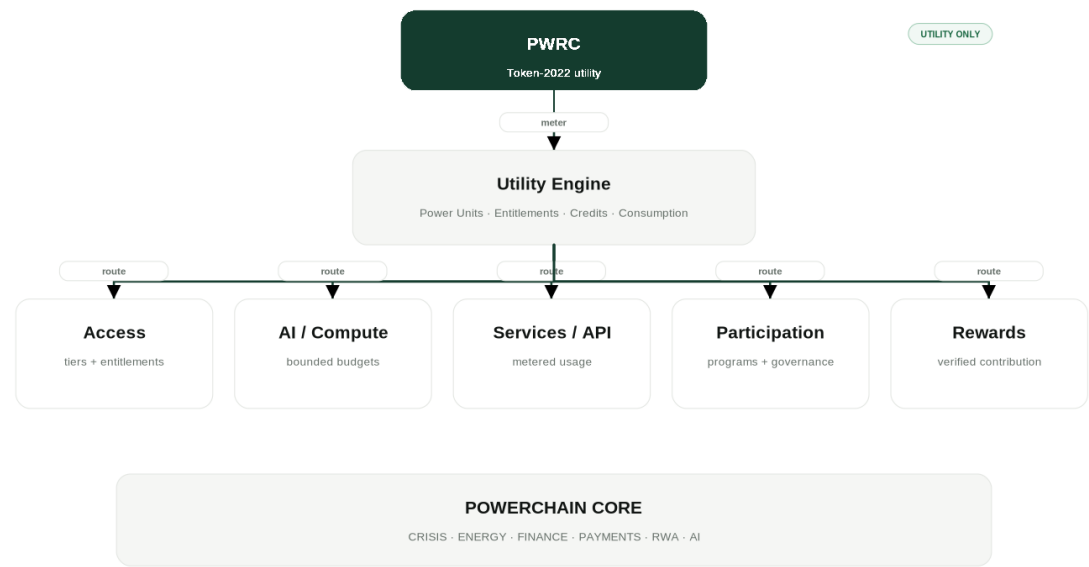
EXECUTIVE SUMMARY

PowerChain connects capital to outcomes

PowerChain Crisis is a capital-coordination and evidence system for verified emergency response. It separates crisis intelligence, stable-asset capital, response execution, trust, settlement and proof of impact so no single wallet, token, AI model or database becomes the entire control plane.

PWRC UTILITY ARCHITECTURE · v1.0.0

Utility powers the network. Stable assets fund relief.



Source: PWRC Utility Architecture v1.0.0 · PWRC balance never grants treasury authority.

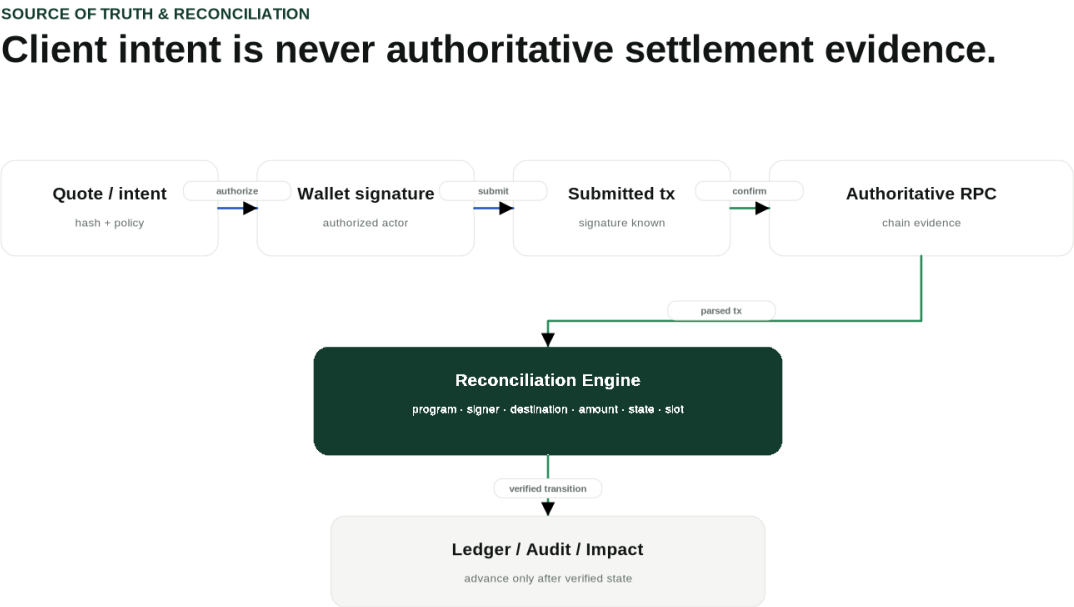
Figure 1. PWRC is a network utility layer, not crisis donor capital.

Challenge	PowerChain control
Slow / fragmented funding	Solana settlement, checkout, Solana Pay and portable Actions/Blinks.
Misallocation risk	Purpose-bound pools, allocation policy, escrow and approval thresholds.
Weak transparency	Explicit financial states and public capital-to-impact views.
Procurement risk	Supplier controls, explainable integrity signals and evidence.
Difficult verification	Evidence Fabric, field verification and Impact Passports.
AI overreach	AI may recommend; policy and authorized signers control irreversible actions.

ARCHITECTURE

Five connected networks inside CCN

The Crisis Capital Network is a network of bounded domains. The product UI can unify them, but the data and authority models remain separate.



Source: PowerChain reconciliation architecture v1.0.0 · Unknown outcome remains EXECUTION_UNKNOWN until reconciled.

Figure 2. Source-of-truth routing separates Solana settlement, external observations, operational data and evidence before state reaches the UI.

Network	Primary responsibility
Crisis	Canonical crisis identity, severity, geography, lifecycle and verified need.
Capital	Pools, capital gap, allocations, escrow and funding intents.
Response	Organizations, suppliers, procurement, logistics and field operations.
Trust	Identity, evidence, policy, risk, verification, attestations and audit.
Settlement	Solana programs/tokens, Actions/Blinks and provider-specific cross-chain settlement.

Truth layers

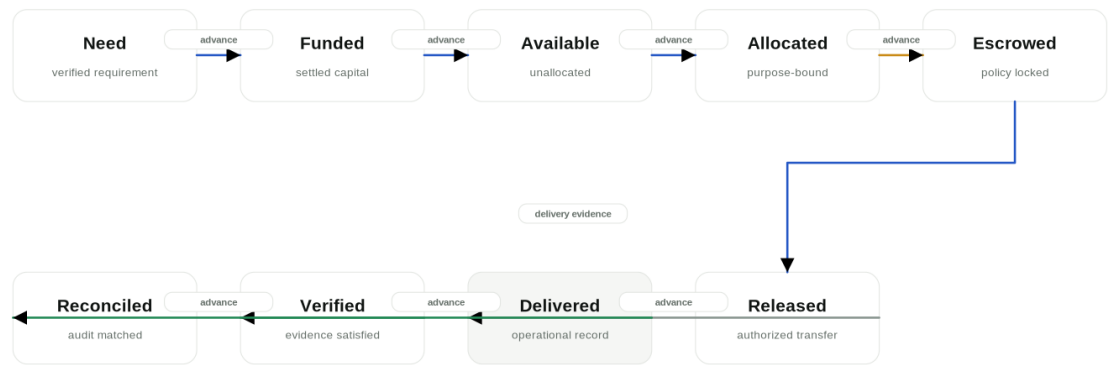
- Solana is authoritative for blockchain settlement and program/token state.
- PostgreSQL/Supabase is authoritative for operational workflows, organizations, policies, evidence metadata and audit indexes.
- Private object storage holds raw sensitive evidence.
- The event engine normalizes authoritative chain/application events for ledger, risk, notifications and analytics.

CAPITAL CONTROL

Financial states remain distinct

CAPITAL LIFECYCLE · DISTINCT STATES

Raised is not released. Released is not verified impact.



Source: CCN capital lifecycle v1.0.0 - State values must never be collapsed into one generic progress percentage.

Figure 3. Capital progresses through explicit states; arrows represent allowed lifecycle transitions rather than decorative flow.

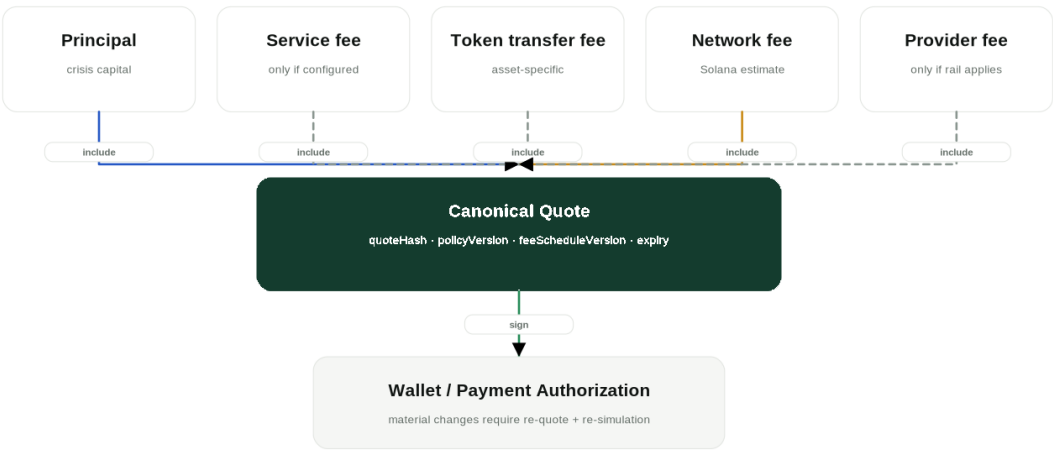
State	Meaning
NEED	Verified capital requirement.
COMMITTED	Capital promised/reserved by an accepted source.
AVAILABLE	Settled capital available to allocate.
ALLOCATED	Budget assigned to a permitted purpose/recipient class.
ESCROWED	Purpose-bound capital reserved under release conditions.
RELEASED	Authorized blockchain/payment movement completed.
DELIVERED	Operational delivery record exists.
VERIFIED	Required evidence/verification threshold is satisfied.
RECONCILED	Financial and operational records agree.

FEE MODEL

Show the entire fee stack before authorization

FEE STACK · SIGNED QUOTE

Every applicable fee is named before authorization.



Source: PowerChain fee model v1.0.0 · The 5% success commission is charged from successful pool proceeds, not contributor checkout.

Figure 4. Material fee and policy fields are bound into the canonical quote.

The contributor checkout shows principal and each applicable service, token-transfer, network or provider fee separately. The 5% successful-funding commission is not a contributor checkout charge: where a capital-pool policy enables it, it is charged from successful pool proceeds only after the authoritative success threshold is satisfied.

Successful-funding commission

Destination	Share of successful pool proceeds
Community Treasury	2 percentage points
Ecosystem & Development	3 percentage points
Total	5 percentage points

ATOMIC SETTLEMENT

Where the asset/network supports it, the 2% and 3% legs should be signed in one transaction and reconciled before the commission record becomes CONFIRMED.

PowerChain Network Utility

PWRC is the shared utility asset for PowerChain access, AI/API consumption, participation, reviewed incentives and economic governance. A PWRC balance does not confer crisis-treasury or emergency authority.

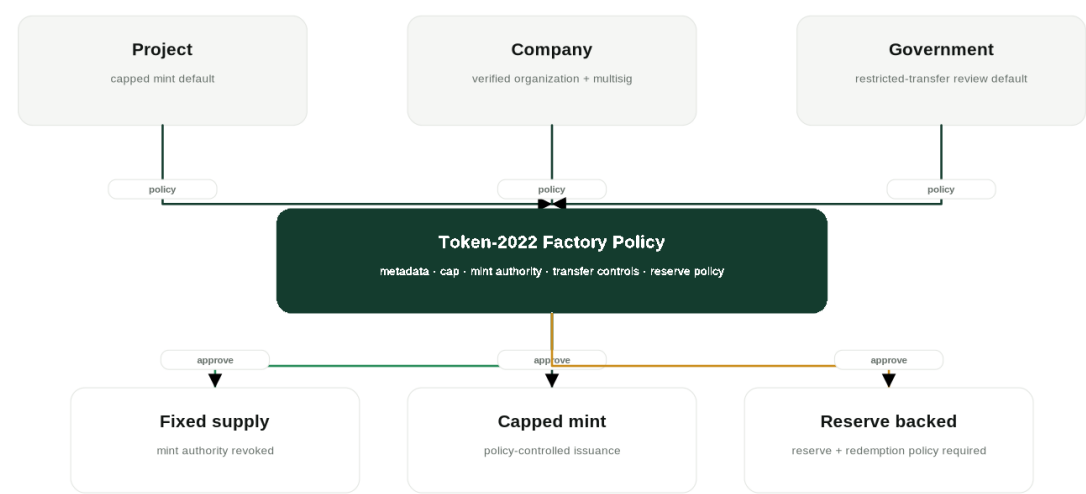
Utility category	Examples
ACCESS	Entitlements and service tiers.
COMPUTE	AI workloads and bounded agent utility budgets.
SERVICES	Reports, data products and selected platform services.
PARTICIPATION	Programs and ecosystem coordination.
GOVERNANCE	Economic governance; not emergency treasury authority.
INCENTIVES	Evidence-backed community/ecosystem rewards.
REPUTATION	One input to eligibility/trust, never the sole score.
NETWORK	API and machine-service metering.

Power Units

Power Units are an internal metering abstraction. The reference planning conversion is configurable; service pricing can change without altering token decimals or crisis-capital accounting.

One factory with safer issuer profiles

TOKEN FACTORY · POLICY FIRST
One factory. Safer defaults by issuer profile.



Source: Token Factory Security v1.0.0 - Default planning cap and fee never bypass issuer, signer, reserve or compatibility review.

Figure 5. Project, company and government assets share the same factory but use different policy defaults.

Profile	Safer default posture
PROJECT	Capped mint, verified issuer, purpose/metadata disclosure.
COMPANY	Verified organization, multisig/HSM policy, explicit treasury/mint separation.
GOVERNMENT	Restricted-transfer review by default and stronger change control.
RESERVE BACKED	Reserve asset, ratio, attestation, redemption and controlled mint/burn policy required.

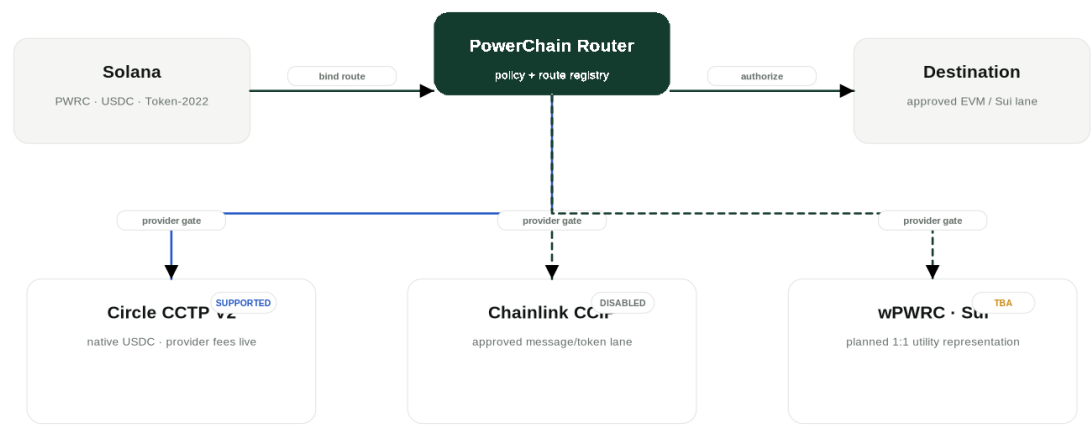
MINT SECRET BOUNDARY

The backend should prepare and verify Token-2022 transactions, but it should not receive or persist the mint private key. A one-billion-unit value is a planning cap, not an automatic mint.

CROSS-CHAIN

Provider-specific routes fail closed

CROSS-CHAIN TRUST ROUTES
No generic bridge switch. Each route has its own controls.



Source: PowerChain cross-chain security model v1.0.0 - Route limits, replay key, finality and audit status are mandatory before execution.

Figure 6. CCTP, CCIP and wPWRC use different trust and reconciliation models.

Route	Purpose	Activation rule
CCTP V2	Native USDC cross-chain movement.	Domains/version/fees/status verified at execution time.
CCIP	Approved messaging/token lanes.	Router/pool/lane, allowlists, rate limits and audit required.
wPWRC / Sui	Planned 1:1 PWRC utility representation.	TBA until two-way bridge audit, custody and supply reconciliation are complete.

Conservation invariant

Outstanding wPWRC must never exceed verified backing PWRC. Any backing deficit or unknown source-chain state should pause new wrapped minting until reconciliation resolves the discrepancy.

Identity, policy and signer authority remain separate

Authentication is not authorization. Token ownership is not authorization. AI output is not authorization.

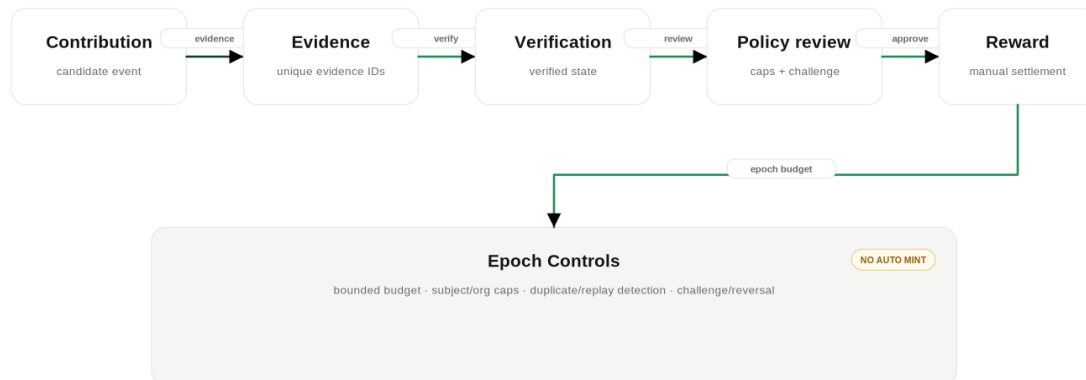
Layer		Control
Identity		Authenticated user/wallet/service identity with replay-resistant challenge/session handling.
Organization		Tenant and verified organization membership.
Role / permission		Least-privilege operating scope and separation of duties.
Policy version		Machine-readable limits, allowed purpose/recipient/asset/network and approvals.
Risk		Explainable rules/anomaly signals; high-risk paths may hold/review.
Approval		Human or governance threshold appropriate to the action.
Signer		Wallet, multisig, HSM/MPC or approved signing infrastructure.
Reconciliation		Authoritative blockchain/provider verification before state advances.
HARD INVARIANT No AI agent, client response, PWRC balance or static configuration can bypass treasury policy or cryptographic signing.		

COMMUNITY REWARDS

Bounded rewards for verified contribution

COMMUNITY REWARD SYSTEM

Reward verified contribution, not passive holding.



Source: Community Rewards roadmap v1.0.0 - Rewards never grant treasury authority or automatically change trust status.

Figure 7. Rewards are evidence- and policy-reviewed, not passive-holding emissions.

Epoch budget model

- budget per epoch;
- per-subject and per-organization caps;
- evidence IDs and duplicate/replay checks;
- conflict-of-interest and anomaly review;
- challenge/reversal procedure;
- separate reward and crisis-capital ledgers.

AUTHORITY BOUNDARY

Reward eligibility does not grant treasury authority, procurement access, verifier status or an automatic reputation upgrade.

EVIDENCE & IMPACT

A transaction is not an outcome

PowerChain closes the lifecycle with field evidence, verification and outcome records. Sensitive raw evidence remains private/off-chain by default; public transparency uses approved metadata, hashes, attestations and human-readable records.

Stage	Control
Upload intent	Organization/crisis/subject binding, MIME/size policy and visibility class.
Storage	Private evidence bucket by default; public media uses a separate class.
Provenance	Server verifies stored object hash before evidence becomes AVAILABLE.
Verification	Verifier/policy records VERIFIED, PARTIAL, PENDING, CONFLICTING, EXPIRED or REJECTED.
Attestation	Optional on-chain hash/reference without publishing private content.
Impact	Outcome records link back to allocation, procurement, delivery and evidence.
Passport	Public or contributor-facing trace from source to verified impact.

Public simplicity, operator control, auditor evidence

Audience	Primary experience
Public	What happened? Can I trust this? Where is capital going? What has been verified?
Contributor	Amount, purpose, asset, network, fee stack, destination, authorization and receipt.
Operator	Alerts, approvals, blocked capital, procurement exceptions and verification backlog.
Treasury	Balances by state, allocations, escrow, releases, signers and reconciliation.
Auditor	Source, policy, approval, transaction, recipient, document, evidence and impact chain.

Visual language

- White / light gray / near-black / dark forest green.
- No neon or decorative crypto effects.
- Color plus text/icon for every status.
- Source and freshness labels on live data.
- Technical transaction details are available but not forced into the primary public flow.
- Mobile prioritizes understand -> trust -> fund -> track -> impact.

Nepal Flood Response - demo only

DEMO DATA	
The figures on this page are product fixtures. They are not real fundraising totals, verified humanitarian claims or live field outcomes.	
Metric	Demo value
Raised	\$4.28M
Allocated	\$3.71M
Verified Impact	\$2.94M
In Escrow	\$770K
Verification rate	72%

Purpose breakdown

Purpose	Demo allocation
Water	\$1.20M
Medical	\$1.70M
Shelter	\$900K
Logistics	\$480K

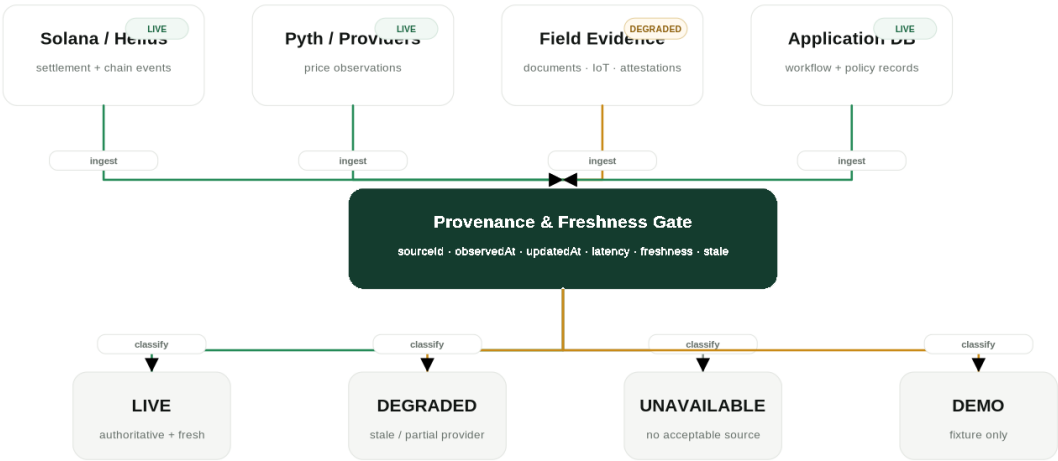
Mainnet/public production pages must either display authoritative live data with source identity/freshness or show an explicit unavailable/degraded state. Demo fixtures must never silently substitute for live crisis data.

Freshness and provenance are first-class

Pricing and external data observations carry provider identity, source, publish timestamp, fetch timestamp, freshness and status. The system must not invent a price or treat an external report as a verified crisis fact.

DATA PROVENANCE & FRESHNESS

Every operational value carries source identity and state.



Source: PowerChain data contract v1.0.0 - Mainnet renders authoritative data or explicit DEGRADED / UNAVAILABLE; demo fixtures stay labeled DEMO.

Figure 8. Provider observations pass through provenance and freshness gates before they can be labeled live, degraded, stale or unavailable.

Provider / layer	Role
Pyth	Primary market-price observation where configured; freshness/confidence enforced.
Birdeye	Corroboration or degraded fallback according to policy.
Chainlink	Provider-specific data/CCIP boundary where configured.
Helius	Solana RPC/API/webhook event infrastructure; webhook events are normalized/idempotent.
PowerChain backend	Policy evaluation, reconciliation, evidence metadata, risk and operational state.

Provider state

Use LIVE, DEGRADED, STALE, DISABLED, TBA or UNAVAILABLE. Static configuration alone does not justify a LIVE label, and unavailable data must never be rendered as a numeric zero.

Lines, arrows and charts carry operational meaning

The architecture and chart system uses one semantic grammar across website, dashboard, docs and exports. Connectors are not decorative: their style communicates execution, optionality, review, degradation and source confidence.

Visual primitive	Required semantics
Node state	Operational state of the entity itself: live, verified, degraded, paused, TBA or unavailable.
Edge state	Independent transition state such as authorize, submit, confirm, release, evidence, reconcile, optional or blocked.
Arrowhead	Direction of authority, data or capital movement. Every process connector terminates explicitly.
Dashed connector	Optional, degraded, disabled or planned route - never a normal confirmed execution path.
Line chart	Latest value, previous-point delta, source identity, update/observation time and point-quality flags.
Point quality	VERIFIED, ESTIMATED, STALE or MISSING; missing data is not plotted as zero.
Source label	Provider plus freshness/latency where relevant. Demo/static values must be visibly labeled.
Accessible fallback	Charts and architecture maps expose a text/table representation for keyboard and screen-reader users.

Mainnet rendering rule

SOURCE BEFORE STYLE Mainnet UI may render an operational value only when an authoritative source and acceptable freshness are available. Otherwise show an explicit DEGRADED, STALE or UNAVAILABLE state. Demo fixtures remain labeled DEMO.

Mobile behavior

Complex left-to-right flows collapse into vertical sequences on narrow screens. Edge labels remain adjacent to the transition they describe, and state is communicated with text/icon plus color rather than color alone.

PRODUCTION READINESS

Fail closed before mainnet

Gate	Required evidence
Dependencies	Pinned runtime/package manager, committed lockfile and clean frozen install.
Database	Prisma validation/migrations, RLS/storage review, backup/restore test.
Programs	Anchor build/tests, deployment IDs, authority/custody policy and independent review.
Financial	Treasury policy, transaction simulation, exact reconciliation and fee disclosure.
Cross-chain	Provider configuration, limits, replay protection, monitoring, incident runbook and audit status.
Token factory	Issuer approval, metadata, signer flow, reserve review and extension compatibility.
AI	Model/tool permission evaluation, prompt-injection testing and audit trail.
Operations	Monitoring, DR, event replay, degraded/offline modes and on-call procedures.
UNKNOWN EXECUTION If the final blockchain/provider result cannot be established, preserve EXECUTION_UNKNOWN and reconcile later. Never convert uncertainty into success.	

Activation sequence - version remains 1.0.0

01 Core financial close

Audit/deploy core Solana programs, real treasury registry, USDC/Solana Pay and authoritative reconciliation.

02 Evidence close

Production storage/RLS, verifier workflow, impact passport and public transparency.

03 PWRC utility close

Power Units, enterprise utility accounts, AI/API budgets and usage receipts.

04 Token factory close

Issuer policy, secure mint signing, reserve-backed controls and Token-2022 compatibility matrix.

05 Cross-chain close

CCTP production routes, audited CCIP lanes and only then wPWRC/Sui activation.

06 Community reward pilot

Epoch budgets, evidence policy, anti-gaming, review and reversals.

07 Partnership network

Scoped partner credentials, verified organizations and integration contracts.

08 Operational SLOs

RPC/webhook/queue/reconciliation/evidence backlog targets and incident dashboards.

CONCLUSION & REFERENCES

Speed without uncontrolled authority

The architectural goal is not to put donations on a blockchain. It is to make emergency capital traceable, policy-controlled and verifiable from crisis identity to real-world outcome.

POWERCHAIN PROMISE

Move capital where it matters. Control how it is used. Prove what it achieved.

Core invariant

PWRC powers the network around the response. Stable assets fund the response. Identity and policy control authority. Cryptographic signers authorize value movement. Evidence proves real-world outcomes.

Primary technical references

Solana Tokenization: <https://solana.com/docs/tokenization>

Solana Confidential Transfer integration: <https://solana.com/docs/tokens/extensions/confidential-transfer/integration-guide>

Circle CCTP: <https://developers.circle.com/cctp>

Chainlink CCIP: <https://docs.chain.link/ccip>

Pyth Network: <https://docs.pyth.network/>

Helius: <https://www.helius.dev/docs>

Technical status, supported routes, package IDs, mint addresses, fees, provider capabilities and audit/deployment state must be verified against the running environment. This whitepaper is not financial, legal, investment or tax advice.